

KIAN

LINEAR POSITION TRANSDUCER

- *Main Features*

- *Very long life*
- *Infinite resolution*
- *Wipers are made of best quality alloys*
- *Linearity more than 99%*
- *Connection suitable for DIN 43650 (hydraulic connector)*
- *High operating speed*
- *Protection class up to IP64*
- *Ideal for applications on plastic injection machines, vertical and horizontal presses, and on many other applications*



- *Technical Data*

Nominal electrical lengths	100/150/200/300/350 400/450/550/650/750 (Other lengths on request)
Independent linearity	± 0.5 %
Resolution	Infinite – Depends on quality of applied voltage
Repeatability	0.01 mm
Electrical connections	DIN 43650
Displacement speed	< 10 m/s
Protection level	IP 54 – Up to IP 64 on request
Recommended operating wiper current	≤ 1uA / 24V DC
Maximum operating wiper current	10mA / 24V DC
Recommended applicable voltage	24V DC
Maximum applicable voltage	36V DC
Working temperature	-20...+80 °C
Storage temperature	-40...+100 °C
Material for transducer Housing	Anodised aluminium
Thermal coefficient of resistance	-500ppm/°C
Temperature coefficient of output voltage	≤ 1.5ppm/°C
Dielectric strength of internal circuit	> 45 kV
Tolerance on resistance	15%

- *Electrical - Mechanical Data*

Model	100	150	200	300	350	400	450	550	650	750
Electrical length (mm)	115	165	215	315	365	415	465	565	665	765
Standard deviation (% of F.S)	0.2%	0.1%	0.05%							
Resistance	5 kΩ							10 kΩ		
Useful electrical length (mm)	110	160	210	310	360	410	460	560	660	760
Mechanical length ±1 (mm)	110	160	210	310	360	410	460	560	660	760
Case length * ±2 (mm)	150	210	260	360	410	460	510	610	710	810

* Case length is the length of housing, not considering sockets.

- ***Installation Manual***

1. Connect power as shown in figure below.

Notice: For each model, electrical length is 15mm greater than nominal length; but first and last 2.5mm are not calibrated. Therefore, useful electrical length would be 10mm greater than nominal length, as mentioned in the table above.

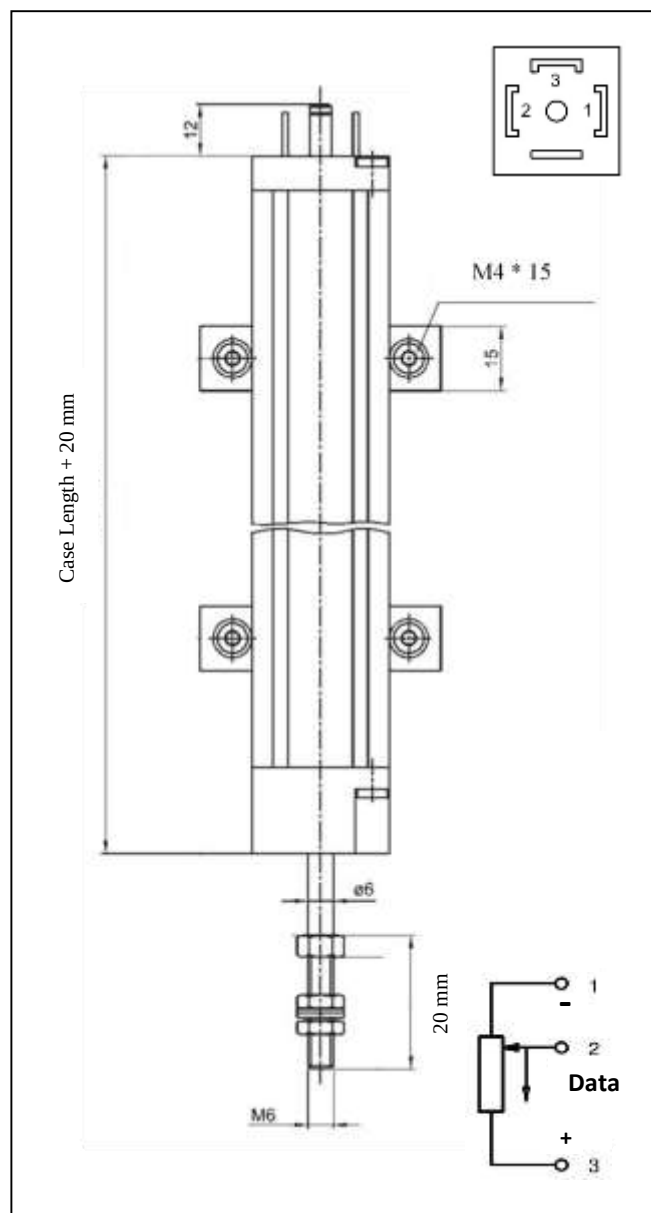
2. According to the above point, completely push the shaft inside, then pull it out 2.5mm and mark the voltage as your **ZERO POINT** voltage. Next pull the shaft completely out, then push it 2.5mm inside. Voltage on this point is your **END POINT** voltage.

3. Transducer output, in mm, is calculated as below:

$$\text{Output (mm)} = k \times (\text{Useful electrical length})$$

$$\text{where } k = \frac{\text{Pin 2 Voltage (v)} - \text{ZERO POINT (v)}}{\text{END POINT (v)} - \text{ZERO POINT (v)}}$$

Your transducer is now calibrated!



** KIAN CO. reserves the right to make any kind of design or functional modification at any moment without prior notice.

- ***Contact Information***

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